计算机科学与技术学院
2024级计算机科学与技术专业（全英文班、留学生）
人才培养方案

Undergraduate International Education Program in Computer Science and Technology，Grade 2024，School of Computer Science and Technology

一、培养目标（Goals）

立足中国、面向全球，培养了解当代中国政治、经济、社会和文化，具有国际视野和团队合作精神，掌握扎实、宽广的计算机专业理论知识和较强的工程实践能力，富有创新思维的复合型工程技术人才，毕业生可在计算机相关领域的产业或科研部门从事计算机理论与应用研究、产品设计与开发，及信息化项目管理等工作，并有能力继续攻读计算机科学与技术及相关学科、交叉学科的硕博士学位。

毕业生在毕业五年内应达到以下目标：

目标1：掌握扎实的自然科学基础知识，掌握计算机学科的基本理论和方法；

目标2：了解当代中国政治、经济、社会和文化，具有良好的人文素养，具备良好的职业道德，能够运用环境、法律、伦理、道德等非技术因素指导研究工作与应用实践；

目标3：具有计算机软硬件系统及其应用系统的设计、开发、集成能力；

目标4：具有综合运用计算机理论、项目管理方法，组织和实施计算机软硬件项目的能力，具备在项目中进行沟通、协调、团队合作的能力；

目标5：具有良好的中英文交流沟通、文档写作能力，具有广阔的国际视野；

目标6：掌握文献检索和资料查找能力，能够持续跟踪和学习计算机领域的前沿技术，具备良好的知识更新能力，并能在一定程度上进行技术创新。

To meet the talents requirement from China and even globe, the undergraduate program in Computer Science is targeted at cultivating middle or higher-level talents with good qualities in humanities and social sciences, international vision, team-working spirit, solid knowledge foundation, good engineering abilities and innovative capacity, who can be prepared for any professional role like research, design, development or management in information technology industries, research institutes, and other community groups, and also for their advanced degree programs in computer science and technology, other relevant fields and cross disciplines.

The skill profile is as following.

1.To gain sound grounding in both mathematics and physics knowledge and principles in computer science;

2. To gain knowledge of humanities and social science, humanistic qualities, good professional ethics and strong sense of responsibility and mission of profession, society and environment for guiding research and engineering practice;

3. To be capable of designing, developing and integrating computer hardware software systems and their applications;

4. To capable of applying the knowledge and methodologies from computer science, software project management, relevant policies and regulations to organize and carry out hardware or software projects, with plenty of harmonizing, managing, communication and cooperating skills;

5.To gain the ability in reading and writing technology archives in Chinese and English language, and an extended global vision;

6. To gain the ability of literature searching, data querying and thesis writing, to keep learning cutting edge and new knowledge in computer science and boost technology innovation.

二、毕业要求（Specification）

毕业要求 1（工程知识）：能够将数学、自然科学、工程基础和专业知识用于解决计算机学科领域问题。

Engineering Knowledge： Be able to use mathematical natural science, engineer fundamentals, and the major knowledge of computer science to solve complex problems in the information field.

毕业要求 2（问题分析）：能够应用数学、自然科学和工程科学的基本原理和方法，识别、表达、并通过用户调研和文献研究分析问题，以获得有效结论。

Problem analysis: Be able to use basic principles of mathematics, natural science, and engineering to recognize and express problems, and analyze the complex problems in computer science through literature research to make effective conclusions.

毕业要求 3（设计/开发解决方案）：能够设计针对计算机领域科学技术问题的解决方案，设计满足特定需求的软硬件系统，并能够在设计和开发环节中体现创新意识，考虑社会、健康、安全、法律、文化以及环境等因素。

Design/Develop solution schema：Be able to give the solution scheme for complex problems in information areas, and the scheme should meet the needs of specific systems, unit (components). The social, health, security, law, literature, and environment factors should be considered in the scheme.

毕业要求 4（研究）：能够基于科学原理并采用科学方法对计算机领域复杂问题进行研究，包括建立模型、设计算法和实验、分析与解释数据，并通过信息综合得到合理有效的结论。

Research：Be able to carry out research for complex engineering problems in the computer science area based on scientific principles and scientific methods, including the design of the experiments, analyze and explain data, and given reasonable and effective conclusions from information fusion.

毕业要求 5（使用现代工具）：能够针对计算机领域复杂问题，开发、选择与使用恰当的技术、资源和工具，包括对问题的预测与模拟，并能够理解其局限性。

Utilize modern tools： Be able to develop, select, and use suitable techniques, resource, and modern engineering tools and information technique tools for complex problems in the computer science area, including the prediction and simulation of the complex problems and understand their limitations.

毕业要求 6（工程与可持续发展）：在解决复杂工程问题时，能够基于软件工程相关背景知识进行合理分析，评价软件工程实践和复杂工程问题解决方案对健康、安全、环境、法

律以及经济和社会可持续发展的影响，并理解应承担的责任。

Engineering and the society: Be able to reasonably analyze and evaluate the influence of the engineering practice and solution schemes to complex problems in computer science on human health, security, environment, law, economy and society's sustainable development , and understand the corresponding responsibilities.

毕业要求 7（伦理和职业规范）：有工程报国、工程为民的意识，具有人文社会科学素养、社会责任感，能够理解和应用工程伦理，在工程实践中遵守工程职业道德、规范和相关法律，履行责任。

Professional morality： Obtain the personal characteristics of having human literature, and social responsibilities, and be able to understand and comply with professional morality and regulations in the engineering practices, and take the corresponding responsibilities.

毕业要求 8（个人和团队）：能够在多样化、多学科背景下的计算机项目研发团队中承担个体、团队成员以及负责人的角色。

Person and team： Be able to take the role of single person, team member, or the team leader in the team of multiple disciplines.

毕业要求 9（沟通）：能够就计算机领域复杂工程问题与业界同行、团队成员和客户进行有效沟通和交流，包括撰写报告和文档、陈述发言、清晰表达或回应指令。并具备一定的国际视野，能够在跨文化背景下进行沟通和交流，理解、尊重语言和文化差异。

Communication： Be able to effectively communicate with colleagues or the public about complex problems in computer areas, including writing and designing reports, making presentation, clearly expressing and responding commands. Moreover, be able to have international view and can communicate and express cross different cultures.

毕业要求 10（项目管理）：理解并掌握计算机项目管理原理与经济决策方法，并能在多学科环境中应用。

Project management： Understand and master the principles of computer science and the methods for economy decision, and can apply them in multi-discipline environments.

毕业要求 11（终身学习）：具有自主学习和终身学习的意识和能力，能够理解广泛的技术变革对工程和社会的影响，适应新技术变革，具有批评性思维能力。

life-long learning: Have the idea and the habit of self-learning and life-time learning, and the abilities of non-stop learning and making progress.

表 2-1　毕业要求指标点分解

Table 2-1　Graduation requirements index

毕业总要求		毕业要求分指标	
编号	内容	指标点	内容
1	工程知识 Engineering Knowledge	1-1	掌握基本的数学、物理基础学科知识和计算机理论知识，并能用于表述计算机领域复杂科学与工程问题。 Master the basic knowledge of mathematics, physics and computer theory, and can be used to express complex scientific and engineering problems in the field of computer
		1-2	能将数学、物理基础学科知识和计算机理论知识用于数学模型的建立并求解，以及算法的分析与设计。

毕业总要求		毕业要求分指标	
编号	内容	指标点	内容
			Apply the basic subject knowledge of mathematics, physics and computer theory to the establishment and solution of mathematical models, as well as the analysis and design of algorithms
		1-3	能用工程基础与计算机理论知识比较和综合计算机领域复杂工程问题解决方案。 Compare and synthesize the solution of complex engineering problems in the field of computer with the basic engineering and computer theory knowledge
2	**问题分析 Knowledge analysis**	2-1	能够应用数学、自然科学和计算机科学的第一性原理，识别和判断计算机领域复杂工程问题的关键环节。 apply the first principles of mathematics, natural science and computer science to identify and judge the key links of complex engineering problems in the computer field
		2-2	能够应用相关科学原理和数学模型方法，正确表达计算机领域复杂工程问题，建立相关模型。 apply relevant scientific principles and mathematical model methods to correctly express complex engineering problems in the computer field and establish relevant models
		2-3	能够运用基本原理，借助文献研究，寻求可替代的解决方案，并从可持续发展的角度分析科学研究和工程实践活动过程的影响因素，获得有效结论。 By using basic principles and literature research, seek alternative solutions, analyze the factors affecting the process of scientific research and engineering practice from the perspective of sustainable development, and obtain effective conclusions
3	**设计/开发解决方案 Design& develop solutions**	3-1	掌握设计、开发计算机领域复杂工程问题解决方案的基本方法和技术，了解影响设计目标和技术方案的各种因素。 Master the basic methods and techniques of designing and developing solutions to complex engineering problems in the computer field, and understand various factors affecting design objectives and technical solutions
		3-2	设计并实现满足特定需求的算法模块、软件系统或硬件单元/部件。 Design and implement algorithm modules, software systems, or hardware units/components that meet specific needs
		3-3	在设计计算机领域科学研究与工程问题的解决方案过程中考虑社会、健康、安全、法律、文化以及环境等因素的影响。 Consider social, health, safety, legal, cultural and environmental factors in the design of solutions to scientific research and engineering problems in the computer field
4	**问题研究 Problem research**	4-1	基于科学原理，通过文献研究等方法，对计算机领域复杂工程问题进行调研和分析，并提出解决方案。 Base on scientific principles, through literature research and other methods, the complex engineering problems in the computer field are investigated and analyzed, and solutions are proposed
		4-2	针对问题本质，明确研究路线，设计实验方案。 According to the essence of the problem, the research route is clear and the experimental scheme is designed
		4-3	能够根据实验方案构建实验系统，收集实验数据，分析实验结果，并通过信息综合得到问题的有效结论。 Built experimental systems according to the experimental scheme, the experimental data is collected, the experimental results are analyzed, and the effective conclusions are obtained through information synthesis
5	**使用现代工具 Using modern toolkit**	5-1	了解计算机专业常用的技术、资源、软件和工具的使用原理和方法，并理解其局限性。 Understand the principles and methods of using techniques, resources, software and tools commonly used in the computer science profession and understand their limitations
		5-2	能够选择与使用恰当的技术、资源、工程工具和专业软件，用于计算机领域复杂科学与工程问题的分析与设计。 select and use appropriate technologies, resources, engineering tools, and specialized

毕业总要求		毕业要求分指标	
编号	内容	指标点	内容
			software for the analysis and design of complex scientific and engineering problems in the computer field
		5-3	能够针对具体的对象，开发或选用满足特定需求的工具，用于模拟和预测专业问题，并能够分析其局限性。 Develop or select tools that meet specific needs for specific objects, for modeling and predicting professional problems, and for analyzing their limitations
6	**工程与可持续发展 Engineering and sustainable development**	6-1	了解计算机专业相关领域的技术标准体系、知识产权、产业政策和法律法规，理解健康、安全、环境、法律以及经济和社会可持续发展等因素对计算机科学研究工程活动的影响。 Understand the technical standards system, intellectual property rights, industrial policies and laws and regulations related to the field of computer science, and understand the impact of health, safety, environmental, legal, and economic and social sustainable development factors on computer science research and engineering activities
		6-2	能够分析与评价计算机专业工程实践和复杂工程问题解决方案对健康、安全、环境、法律以及经济和社会可持续发展的影响，以及这些制约因素对项目实施的影响，并理解应承担的责任。 analyze and evaluate the health, safety, environmental, legal, and economic and social sustainability impacts of computer engineering practices and solutions to complex engineering problems, as well as the impact of these constraints on project implementation
7	**职业规范** Profession norm	7-1	具有人文社会科学素养，树立正确的世界观、人生观、价值观。 With humanities and social science literacy, establish a correct world outlook, outlook on life, values
		7-2	理解计算机领域的工程职业道德规范和社会责任，并能在工程实践中自觉遵守和履行。 Understand engineering ethics and social responsibility in computer field, and be able to consciously abide by and fulfill in engineering practice
8	**个人和团队** Individual and team	8-1	拥有健康的体魄、良好的心理素质。 Have a healthy physical body and good psychological quality
		8-2	有团队合作精神，能与其他学科的成员合作共事，在团队中独立或合作开展工作。 Be a team player, able to work with members of other disciplines and work independently or cooperatively in a team
		8-3	具有一定的管理能力，能够在多学科背景下的项目团队中承担负责人角色。 Have management skills and be able to assume the role of leader in a multi-disciplinary project team
9	**沟通 Communication**	9-1	具备在科学研究、工程实践等活动中与他人进行沟通和交流的基本能力和素养，理解与业界同行和社会公众交流的差异性。 Have the basic ability and quality to communicate and communicate with others in scientific research, engineering practice and other activities, and understand the differences in communication with industry peers and the public
		9-2	能够就计算机领域复杂科学与工程问题进行有效沟通和交流，通过撰写报告、设计文稿、陈述发言等方式清晰表述问题和方案，并回答质询。 Effectively communicate and exchange complex scientific and engineering problems in the field of computer, clearly express problems and solutions by writing reports, designing documents, presentations and other means, and answer questions
		9-3	具备国际化视野，能跨语种、跨文化就专业问题进行交流和学习。 With an international perspective, able to communicate and learn on professional issues across languages and cultures
10	**项目管理 Project management**	10-1	理解并掌握工程管理原理和经济决策方法，具备初步的项目管理能力。 Understand and master the principles of project management and economic decision-making methods, with preliminary project management ability
		10-2	能在多学科环境下，选择和运用适当的管理和经济决策方法进行计算机研究课题或工程项目的组织和实施。

毕业总要求		毕业要求分指标	
编号	内容	指标点	内容
			Select and apply appropriate management and economic decision-making methods for the organization and implementation of computer research topics or engineering projects in a multidisciplinary environment
11	**终身学习 Life-long learning**	11-1	能够正确认识终身学习的重要性，具有终身学习的意识。 Be able to correctly understand the importance of lifelong learning and have the consciousness of lifelong learning
		11-2	掌握不断学习和适应发展的方法和技能。 Master the methods and skills to continuously learn and adapt to development

三、“培养目标-毕业要求”和“毕业要求-课程体系”支撑矩阵

表 3-1 专业毕业要求对培养目标的支撑矩阵

Table 3-1 Major graduation specification vs. goals correlation matrix

目标 / 要求	**培养目标 1 Goal 1**	**培养目标 2 Goal 2**	**培养目标 3 Goal 3**	**培养目标 4 Goal 4**	**培养目标 5 Goal 5**	**培养目标 6 Goal 6**
毕业要求 1 Specification 1	√					
毕业要求 2 Specification 2	√					
毕业要求 3 Specification 3			√			
毕业要求 4 Specification 4			√			
毕业要求 5 Specification 5			√			√
毕业要求 6 Specification 6				√		
毕业要求 7 Specification 7		√		√		
毕业要求 8 Specification 8					√	
毕业要求 9 Specification 9		√			√	
毕业要求 10 Specification 10				√		
毕业要求 11 Specification 11						√

表 3-2 专业课程体系（必修课程）对毕业要求的支撑矩阵

Table 3-2 Syllabus vs. major graduation specification correlation matrix

课程名称 Course name	毕业要求 1 Specifition1			毕业要求 2 Specifition2			毕业要求 3 Specifition3			毕业要求 4 Specifition4			毕业要求 5 Specifition5			毕业要求 6 Specifition6		毕业要求 7 Specifition7		毕业要求 8 Specifition8			毕业要求 9 Specifition9			毕业要求 10 Specifition10		毕业要求 11 Specifition11	
	①	②	③	①	②	③	①	②	③	①	②	③	①	②	③	①	②	①	②	①	②	③	①	②	③	①	②	①	②
汉语（一） Chinese I																		H							H			L	
汉语（二） Chinese II																		H							H			L	
始业教育 Initial Education																M		H	H										
当代中国 Contemporary China																M		H	H										
中国概况 Chinese Overview																M		H	H									L	
中国国情系列 Chinese National Conditions course series																M		H	H										
大学体育 college physical education																				H	H								
大学外语 College English																							H	M	H				
计算机科学导论 Introduction to Computer Science													H			H		H										M	
C 语言程序设计 ★ C Programming Language	H	H					H						M																L
线性代数 Linear Algebra		H		H																									L
高等数学 B（一） higher mathematics B I		H		M																								H	L
数据结构★ Data Structure and Algorithms		H	H		M		H						H																
高等数学 B（二） higher mathematics B II		H		M																								H	L
大学物理 D University Physics D	H												H															H	

课程名称 Course name	毕业要求 1 Specifition1			毕业要求 2 Specifition2			毕业要求 3 Specifition3			毕业要求 4 Specifition4			毕业要求 5 Specifition5			毕业要求 6 Specifition6		毕业要求 7 Specifition7		毕业要求 8 Specifition8			毕业要求 9 Specifition9			毕业要求 10 Specifition10		毕业要求 11 Specifition11	
	①	②	③	①	②	③	①	②	③	①	②	③	①	②	③	①	②	①	②	①	②	③	①	②	③	①	②	①	②
大学物理实验 D University Physics Experiments D											H	M																	
概率与数理统计 Probability and Statistics		H		M																						H			H
计算机组成与结构★ Computer Organization and Architecture			H	L				M	H																				
离散数学★ Discrete Mathematics	H			H																									H
计算机网络★ Computer Networks			H	H			H				L					H													
操作系统 ★ Operating System			H	H	H				H																				
人工智能基础 Introduction to Artificial Intelligence											H	H		H			H		H										
专业导论 Introduction to Major													M			M			H									H	
计算机新技术讲座 Lecture on New Computer Technology													M										H		H			M	
Java 程序设计 Java Programming Language							H							H			H												
数据库原理及应用 Database Principle and Its Application			H	H	H	M		H																					
软件工程基础 Fundamentals of Software Engineering			H		H			M									H										L		
算法设计与分析 Algorithm Design and Analysis				H		H	M	H				H																	
面向对象分析与设计 Object-Oriented Analysis and Design				H		H	M	H				H																	
C 语言课程设计 C programming Course Design				H				H				H		H															

课程名称 Course name	毕业要求 1 Specifition1			毕业要求 2 Specifition2			毕业要求 3 Specifition3			毕业要求 4 Specifition4			毕业要求 5 Specifition5			毕业要求 6 Specifition6		毕业要求 7 Specifition7		毕业要求 8 Specifition8			毕业要求 9 Specifition9			毕业要求 10 Specifition10		毕业要求 11 Specifition11	
	①	②	③	①	②	③	①	②	③	①	②	③	①	②	③	①	②	①	②	①	②	③	①	②	③	①	②	①	②
数据结构课程设计 DataStructure Practice				H				H				H		H															
Java 课程设计 Java Programming Course Design				H				H				H		H															
专业见习 Professional Pre-internship										H						H			M		H			H				H	
算法设计与分析课程设计 Algorithms Course Design				H				H				M		H															
计算机项目实训（一）——个人软件项目实训 Computer Project Training I ——Solo Software Project Practice					H			H				H			H														
计算机项目实训（二）——团队项目实训 Computer Project Training II ——Team Project Practice			H			H			H									H			H			M					
计算机项目实训（三）——计算机创意设计实训 Computer Project Training III ——Computer Creative Design Practice			H			H			H									H			H			M					
专业基础技能考核 Professional Skill Exam													M	H															
科技文献检索与论文写作 Literature Retrieval and Paper Writing										H		L							M					H					H
毕业实习 Internship													H		H		H		H								H	H	M
毕业论文（设计） Graduation Dissertation（Design）							H	H	M	H	H	H	H	H	H									H			M		H

三、学制（Length of Program）

学制 4 年，在校学习年限 3-6 年。

The program is a four year undergraduate program，and students can complete the program within 3- 6 years.

四、学分修读要求(Credit Requirement)

课程类型	总学分	各学期学分分配									开课门数	最低修读学分
		1	2	3	4	5	6	7	8	滚动开设		
公共基础课程	37	12	11	6	—	—	—	—	—	8	—	31
通识教育课程	—	—	—	—	—	—	—	—	—	—	—	4
学科基础课程	47.5	14	13.5	10.5	9.5	—	—	—	—	—	14	47.5
专业核心课程	16.5	0.5	—	7.5	5	3.5	—	—	—	—	7	16.5
专业方向课程	46	—	2.5	2.5	10.5	12.5	15.5	2.5	—	—	17	11
综合实践课程	35	—	1	1	1	7	2.5	8.5	14	—	12	35
合计	182	26.5	28	27.5	26	23	18	11	14	8	—	145

国际学生毕业时中文能力应当达到《国际汉语能力标准》四级水平。

Course Module	**Total credits**	**Semester schedule**									**Course number**	**Minimum credits**
		1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**rolling**		
General fundamental courses	37	12	11	6	—	—	—	—	—	8	—	31
General education courses	—	—	—	—	—	—	—	—	—	—	—	4
Disciplinary fundamental courses	47.5	14	13.5	10.5	9.5	—	—	—	—	—	14	47.5
Disciplinary core courses	16.5	0.5	—	7.5	5	3.5	—	—	—	—	7	16.5
Disciplinary specialized courses	46	—	2.5	2.5	10.5	12.5	15.5	2.5	—	—	17	11
Practice courses	35	—	1	1	1	7	2.5	8.5	14	—	12	35
Total	182	26.5	28	27.5	26	23	18	11	14	8	—	145

Upon graduation，International students' Chinese language proficiency should reach Level 4 of HSK，the Chinese Proficiency Test.

五、课程设置与安排（**Course Modules and Schedule**）

（一）公共基础课程（General fundamental courses）

课程模块 Course module	修读性质 Course Type	编号课程 course no.	课程名称 course name	学分 credits	周学时/周数 Week education hour/Weeks	总学时 Total education hour	学时分配 Hours Allocated			开课学期 Open semester	备注 Notes
							讲授 Lectures	课程实践 Practice	实验或上机 Experiments or online hours		
汉语类 Chinese	必修 Compulsory	000000X110A101	汉语（一） Chinese I	8	8/16	128	128			1	
		000000X110A102	汉语（二） Chinese II	8	8/16	128	128			2	
	小 计 **Subtotal I**			**16**		**256**	**256**				

课程模块 Course module	修读性质 Course Type	编号课程 course no.	课程名称 course name	学分 credits	周学时/周数 Week education hour/Weeks	总学时 Total education hour	学时分配 Hours Allocated 讲授 Lectures	课程实践 Practice	实验或上机 Experiments or online hours	开课学期 Open semester	备注 Notes
国情教育 National Condition	必修 Compulsory	000000X110A105	始业教育 Initial Education	1	1/16	16	16			1	
		000000X110A103	中国概况 Chinese Overview	3	3/16	48	48			3	
		000000X110A104	当代中国 Contemporary China	3	3/16	48	48			3	
	选修 Selective	详见课程组 Course series	中国国情系列 Chinese National Conditions course series	最少修读 2 学分 Minimum credit: 2							
			大学体育 College Physical ducation	最少修读 4 学分 Minimum credit:4							
	小 计 subtotal III			**13**		**112**	**112**				
其他课程 Other	选修 Selective	详见课程组 Course series	大学外语（一） College English I	3	3/16	48	48			1	
			大学外语（二） College English II	3	3/16	48	48			2	
			大学外语（三） College English(Extended)	2	2/16	32	32			滚动开设	
	小 计 subtotal IV			**8**		**128**	**128**				**至少修读 2 学分 Minimum credit: 2**
合 计 total				**37**		**496**	**496**				**至少修读 31 学分 Minimum credit: 31**

（二）通识课程（General Education Courses）

修读性质 Course Type	课程模块 Course Modules	修读要求 Reading Requirements	备注 Notes
Selective 选修	艺术素养 Art cultivation	具体课程参见总则部分《浙江师范大学通识课程开课计划表》 Courses presented in the university's class schedule	至少修读 4 学分 **Minimum credit: 4**
	社会人文 Social humanity		
	自然科学 Natural science		
	科技应用 Application of science and technology		
	健康生活 Health and life		

（三）学科基础课程（Disciplinary Fundamental Courses）

修读性质 Course Type	编号课程 course no.	课程名称 course name	学分 credits	周学时/周数 Week education hour/Weeks	总学时 Total education hour	学时分配 Hours Allocated 讲授 Lectures	课程实践 Practice	实验或上机 Experiments or online hours	开课学期 Open semester	备注 Notes
必修 Compulsory	080901E130A001	计算机科学导论 Introduction to Computer Science	2	2/16	32	32			1	
	080901E130B001	C 语言程序设计 ★ C Programming Language	3+1	3+2/16	80	48		32	1	
	080901E130A002	线性代数 Linear Algebra	3	3/16	48	48			1	
	000000E130A012	高等数学 B（一） higher mathematics B I	5	6/16	96	96			1	
	080901E130B002	数据结构★ Data Structure and Algorithms	4+1	4+2/16	96	64		32	2	
	000000E130A013	高等数学 B（二） higher mathematics B II	5	6/16	96	96			2	
	000000E130A003	大学物理 D University Physics D	3	3/16	48	48			2	
	000000E130C003	大学物理实验 D University Physics Experiments D	0.5	3/6	18			18	2	
	080901E130A003	概率与数理统计 Probability and Statistics	3	3/16	48	48			3	
	080901E130B003	计算机组成与结构★ Computer Organization and Architecture	3+0.5	3+1/16	64	48		16	3	
	080901E130A004	离散数学★ Discrete Mathematics	4	4/16	64	64			3	
	080901E130B004	计算机网络★ Computer Networks	3+0.5	3+1/16	64	48		16	4	
	080901E130B005	操作系统 ★ Operating System	3+0.5	3+1/16	64	48		16	4	
	080901E130B006	人工智能基础 Introduction to Artificial Intelligence	2+0.5	2+1/16	48	32		16	4	
合 计 Total			**47.5**		866	720		146		

（四）专业核心课程（Disciplinary Core Courses）

修读性质 Course Type	编号课程 course no.	课程名称 course name	学分 credits	周学时/周数 Week education hour/Weeks	总学时 Total education hour	学时分配 Hours Allocated 讲授 Lectures	课程实践 Practice	实验或上机 Experiments or online hours	开课学期 Open semester	备注 Notes
必修 Compulsory	080901E140A001	专业导论 Introduction to The Major	0.5		8	8			1	
	080901E140A002	计算机新技术讲座 Lecture on New Computer Technology	1	4/4	16	16			3	
	080901E140B001	Java 程序设计 Java Programming Language	2+1	2+2/16	64	32		32	3	
	080901E140B002	数据库原理及应用 Database Principle and Its Application	3+0.5	3+1/16	64	48		16	3	
	080901E140B003	软件工程基础 Fundamentals of Software Engineering	2+0.5	2+1/16	48	32		16	4	
	080901E140B004	算法设计与分析 Algorithm Design and Analysis	2+0.5	2+1/16	48	32		16	4	
	080901E140B005	面向对象分析与设计 Object-Oriented Analysis and Design	3+0.5	3+1/16	64	48		16	5	
合 计 total			**16.5**		312	216		96		

（五）专业方向课程（Disciplinary Specialized Courses）

修读性质 Course Type	编号课程 course no.	课程名称 course name	学分 credits	周学时/周数 Week education hour/Weeks	总学时 Total education hour	学时分配 Hours Allocated			开课学期 Open semester	备注 Notes
						讲授 Lectures	课程实践 Practice	实验或上机 Experiments or online hours		
Selective 选修	080901E151B001	Web 前端开发技术 Front-End Web Development	2+0.5	2+1/16	48	32		16	2	
	080901E151B002	UI/UX 交互设计 UI/UX Interactive Design	2+0.5	2+1/16	48	32		16	3	
	080901E151B003	移动应用开发技术 Mobile Application Development	2+0.5	2+1/16	64	48		16	4	
	080901E151B004	Java EE 高级编程 Java EE Advanced Programming	2+1	2+2/16	64	32		32	4	
	080901E151B005	信息安全导论 Introduction to Information Security	2+0.5	2+1/16	48	32		16	4	
	080901E151B006	基于 AI 的创意设计 AI-based Creative Design	2+0.5	2+1/16	48	32		16	4	
	080901E151B007	Python 与数据分析 Python and Data Analysis	2+1	2+2/16	64	32		32	5	
	080901E151A001	软件过程与文档写作 Software Documentation	3	3/16	48	48			5	
	080901E151B008	数字图像处理与计算机视觉 Digital Image Processing and Computer Vision	3+0.5	3+1/16	64	48		16	5	
	080901E151B009	软件质量保证与测试 Software Quality Assurance and Testing	2+1	2+2/16	64	32		32	5	
	080901E151B010	软件体系结构 Software Architecture	2+0.5	2+1/16	48	32		16	6	
	080901E151B011	物联网原理与应用 Internet of Things Principles and Its Applications	2+0.5	2+1/16	48	32		16	6	
	080901E151B012	智能感知技术 Intelligent Sensing Technology	2+0.5	2+1/16	48	32		16	6	
	080901E151B013	大数据技术及应用 Big Data Technologies with Applications	2+0.5	2+1/16	48	32		16	6	
	080901E151A002	软件项目管理 Software Project Management	2	2/16	32	32			6	
	080901E151B014	模式识别与机器学习 Pattern Recognition and Machine Learning	3+0.5	3+1/16	64	48		16	6	
	080901E151B015	防火墙与入侵检测 Firewall and Invasion Detection	2+0.5	2+1/16	48	32		16	7	
合 计 total			46		896	608		288		至少修读 **11** 学分 **Minimun credit：** 11

（六）实践教学课程（Practice Courses）

修读性质 Course features	课程编号 Course no	课程名称 Course name	学分 Credits	总学时 Total education hours	开设学期 Open semester	备注 Notes
必修 compulsory	080901E160C005	C 语言课程设计 C programming Course Design	1	32	2	第二学期前三周，第一学期结束如有短学期，则安排短学期
	080901E160C006	数据结构课程设计 DataStructure Practice	1	32	2 短	
	080901E160E001	专业见习 Professional pre-internship	2	2 周	3 短-4 短	企业认知（3 短） 企业考察（4 短） 实地见习（大二暑假）
	080901E160C007	Java 课程设计 Java Programming Course Design	1	32	4	
	080901E160C008	算法设计与分析课程设计 Algorithms Course Design	1	32	4 短	
	080901E160C001	计算机项目实训（一） —— 个人软件项目实训 Computer Project Training I —— Solo Software Project Practice	2	64	5 短-5	短学期完成 32 学时，另外 32 学时需要在第 5 学期完成
	080901E160C002	计算机项目实训（二） —— 团队项目实训 Computer Project Training II —— Team Project Practice	2	64	5	
	080901E160C003	计算机项目实训（三） —— 计算机创意设计实训 Computer Project Training III —— Computer Creative Design Practice	2	64	6	
	080901E160C004	专业基础技能考核 Professional Skill Exam	0.5	16	6	
	080901E160C009	科技文献检索与论文写作 Literature Retrieval and Paper Writing	0.5	16	6 短	
	080901E160E002	毕业实习 Internship	8	8 周	7	
	080901E160E003	毕业论文（设计） Graduation Dissertation（Design）	14		7、8	
合计			**35**	352		

六、学位课程及修读要求(Degree Courses And Requirements)

1．学位课程名称（列表中带“★”的课程）

C 语言程序设计　　数据结构　　计算机组成与结构

离散数学　　计算机网络　　操作系统

1. Degree Courses(Courses with ★ in the lists above)

C Programming Language　Data Structures　Computer Composition and Structure

Discrete Mathematics　Computer Networks　Operating System

2．学位课程修读要求

学位课程是本专业最核心的课程，是学习其它专业课程的基础。学位课程必须要有准确的课程目标、系统的课程经验、有效的实施方案、科学的评价方式，在充分保证学时学分的前提下，采取各种方式切实提高学位课程教与学的要求与质量，为专业学习打好坚实基础。

2. Degree Course Studying Requirements

Degree Courses are the most important of all, which are the foundation of other major courses. Degree Courses must have clear objectives, detailed plan and scientific evaluating methods in teaching. In guarantee of enough credits and hours, the quality of teaching and studying should be improved by all ways.

七．第二专业课程证书学分要求(Credit Requirements for the Second Major)

修读计算机科学与技术（全英文）第二专业课程证书，须修读本专业下列课程：C 语言程序设计、离散数学、数据结构、操作系统、计算机网络、计算机组成与结构、数据库原理及应用、人工智能基础、Java 程序设计、面向对象分析与设计、算法设计与分析，合计 38.5 学分。

Students who want to get the Second Major Certificate in Computer Science and Technology must achieve a minimum of 38.5 credits by studying the following courses: C Programming Language, Data Structure and Algorithms, Discrete Mathematics, Operating System, Computer Networks, Computer Organization and Architecture, Database Principle and Its Application, Introduction to Artificial Intelligence, Java Programming Language, Object-Oriented Analysis and Design, Algorithm Design and Analysis.